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# SMIC Process Design Kit (PDK)

## Reference Manual

|                     |                                                          |
|---------------------|----------------------------------------------------------|
| <b>PDK Code</b>     | SPDK18MSE_1833_OA_CDS                                    |
| <b>EDA Platform</b> | Cadence                                                  |
| <b>Title</b>        | 0.18um Mixed Signal Enhanced 1.8V/3.3V<br>Cadence OA PDK |
| <b>PDK Revision</b> | V1.11_4                                                  |
| <b>Release Date</b> | 2019-1-15                                                |

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## 3. OVERVIEW

The Purpose of this document is to describe the technical details of the SMIC 0.18um Process Design Kit (PDK) provided by Semiconductor Manufacturing International Corporation. This PDK requires the UNIX environment variable CDS\_Netlisting\_Mode to be set to Analog. Training is not provided as part of this PDK.

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## 4. Supported Design Tools

Cadence Design Framework II, version IC616.500.3 Virtuoso Custom Design Platform:  
Virtuoso Schematic Editor  
Virtuoso Analog Design Environment, including Spectre, AMS, and HspiceD  
Virtuoso Aptivia  
Technology file support for Preview  
VirtuosoLE, VirtuosoXL, Virtuoso CCAR & Chip Editor  
Calibre DRC, LVS, and XRC  
CDL netlisting  
Stream In/Out

## 5. Supported Process Option

Process options can be supported in this PDK:

| PCELL & RULE FILE     | PCELL | DRC<br>(Calibre) | LVS<br>(Calibre) | XRC<br>(Calibre) |
|-----------------------|-------|------------------|------------------|------------------|
| Metallization Options |       |                  |                  |                  |
| 1P6M_5la_1TMa1_MIM10  | V     | V                | V                | X                |
| 1P5M_4la_1TMa1_MIM10  | V     | V                | V                | V                |
| 1P4M_3la_1TMa1_MIM10  | V     | V                | V                | X                |
| 1P3M_2la_1TMa1_MIM10  | V     | V                | V                | X                |
| 1P6M_5la_1TMa1_MIM20  | V     | V                | V                | X                |
| 1P5M_4la_1TMa1_MIM20  | V     | V                | V                | X                |
| 1P4M_3la_1TMa1_MIM20  | V     | V                | V                | V                |
| 1P3M_2la_1TMa1_MIM10  | V     | V                | V                | X                |
| 1P6M_5la_1TMa2_MIM10  | V     | V                | V                | X                |
| 1P5M_4la_1TMa2_MIM10  | V     | V                | V                | X                |
| 1P4M_3la_1TMa2_MIM10  | V     | V                | V                | X                |
| 1P3M_2la_1TMa2_MIM10  | V     | V                | V                | X                |
| 1P6M_5la_1TMa2_MIM20  | V     | V                | V                | X                |
| 1P5M_4la_1TMa2_MIM20  | V     | V                | V                | X                |
| 1P4M_3la_1TMa2_MIM20  | V     | V                | V                | X                |
| 1P3M_2la_1TMa2_MIM10  | V     | V                | V                | X                |
| 1P6M_5la_1TMa3_MIM10  | V     | V                | V                | X                |

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|                      |   |   |   |   |
|----------------------|---|---|---|---|
| 1P5M_4la_1TMa3_MIM10 | V | V | V | X |
| 1P4M_3la_1TMa3_MIM10 | V | V | V | X |
| 1P3M_2la_1TMa3_MIM10 | V | V | V | X |
| 1P6M_5la_1TMa3_MIM20 | V | V | V | X |
| 1P5M_4la_1TMa3_MIM20 | V | V | V | X |
| 1P4M_3la_1TMa3_MIM20 | V | V | V | X |
| 1P3M_2la_1TMa3_MIM10 | V | V | V | X |
| 1P6M_5la_1TMa4_MIM10 | V | V | V | X |
| 1P5M_4la_1TMa4_MIM10 | V | V | V | X |
| 1P4M_3la_1TMa4_MIM10 | V | V | V | X |
| 1P3M_2la_1TMa4_MIM10 | V | V | V | X |
| 1P6M_5la_1TMa4_MIM20 | V | V | V | X |
| 1P5M_4la_1TMa4_MIM20 | V | V | V | X |
| 1P4M_3la_1TMa4_MIM20 | V | V | V | X |
| 1P3M_2la_1TMa4_MIM10 | V | V | V | X |

Notice: “V” means can be supported in this PDK.

“X” means cannot be supported in this PDK.

## 6. What makes up a PDK?

PDK stands for Process Design Kit. A PDK contains the process technology and needed information to do device-level design in the Cadence DFII environment.

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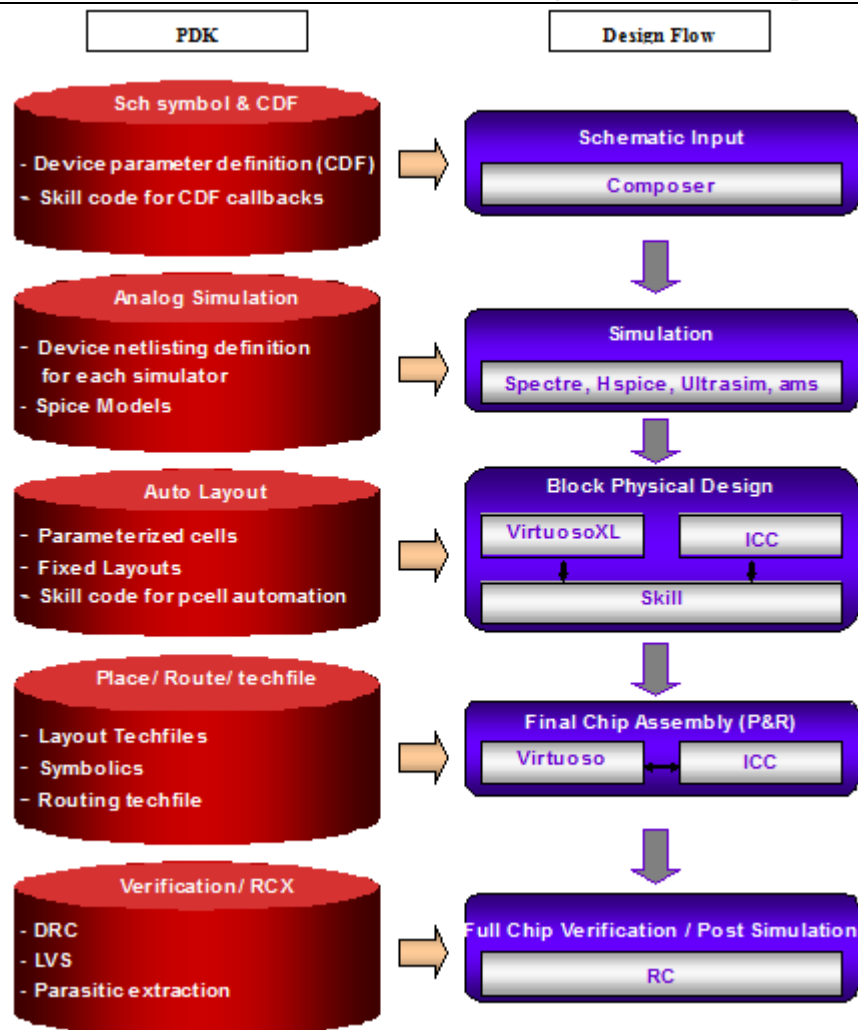


Figure6-1 Design Flow

## 7. Installation of the PDK

### PDK Install Script

To install the PDK, logon to the computer as the user who will own and maintain the PDK.

Choose a disk and directory under which the PDK will be installed. This disk should be exported to all client machines and must be mounted consistently across all client machines.

A perl script ("SMIC\_PDK\_install") which is under the PDK decompression folder (SPDK18MSE\_1833\_OA\_CDS\_Vx.x\_x stands for PDK version) is provided to install SMIC PDK with required metal option (input from user) to a specified folder under installation path.

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```
-b <virtuoso_bin>   Virtuoso tool installation path (Optional)
-h                   Output help text
```

Before using this script, please make sure you can start virtuoso in your working directory, otherwise, please setup

the lcfb installation path using "-b <virtuoso\_bin>" option.

```
For example: ". /SMIC_PDK_install -b
/home/eda/cadence/IC616.500.3 20131102/tools/dfl/bin/virtuoso "
```

The virtuoso version should be IC6.1.6-64b.500.3 or above.

### Example of using the script:

To install 1P6M\_5la\_1TMa1\_MIM10 PDK, execute "SMIC\_PDK\_install" and choose the metal option:

```
./SMIC_PDK_install
```

=====

=====

Virtuoso version should be IC6.1.6-64b.500.3 or above

=====

=====

NOTICE: This PDK only contains RC extraction files for below metal options:

1P6M, 1P5M, 1P4M for topmetal=9k,

1P6M for topmetal=22k, topmetal=34k and topmetal=40k,

For others metal options, user can download RC files from SMICNOW, or raise request if no such supply, SMIC will deliver upon request.

=====

=====

This script will launch /home/eda/cadence/IC616.500.3\_20131102/tools/dfll/bin/virtuoso to set the library.

Please choose 1.8V/3.3V process or pure 3.3V process for installation:

1) 1.8V/3.3V

2) 3.3V

Enter 1 or 2: 1

Please choose the number of top metal for installation:

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1) 1 Top Metal

Enter 1: 1

Please choose the type of 1 Top metal for installation:

1) 9K top metal layer, thickness 9kA (TMa1)

2) 22K top metal layer, thickness 22kA (TMa2)

3) 34K top metal layer, thickness 34kA (TMa3)

4) 40K top metal layer, thickness 40kA (TMa4)

Enter 1, 2, 3 or 4: 1

Please choose the number of inter metal layers:

1) 2 inter metals

2) 3 inter metals

3) 4 inter metals

4) 5 inter metals

Enter 1, 2, 3 or 4: 4

Please choose the type of MIM for installation:

1) MIM1.0fF, Cspec=1fF

2) MIM2.0fF, Cspec=2fF

Enter 1 or 2 : 1

The PDK metal option is: 1P6M\_5la\_1TMa1\_MIM10

Do you want to install yes/no?[yes|no]:yes

PDK installing .....

install DRC.....

install LVS.....

install RC.....

PDK installation complete.

The PDK is ./smic18mse\_1833\_1P6M\_5la\_1TMa1\_MIM10\_oa\_cds\_yy\_mm\_dd\_vx.x\_x

This script will produce a single directory with a name similar to following format:

smic18mse\_1833\_<metal\_option>\_oa\_cds\_yy\_mm\_dd\_vx.x\_x

This is the PDK installation directory for the SMIC 0.18um Mixed Signal Enhanced 1.8v/3.3v Enhanced process.

The default permissions on the PDK have already been set to allow only the owner to have write, read and execute access. Other users will have only read and execute access.



## 8. PDK Install Directory Structure/Contents

cds.lib – Cadence library definition file.

display.drf – Cadence Display Resources File

smicxxxxx/ – SMIC PDK Library

smicxxxxx/techfile.tf – ASCII version of technology file

models/ - directory that will store the PDK models

stream/ - directory containing the cadence stream in and out maps

docs/ - directory containing the SPICE model / Layout techfile / DRC / LVS / Parasitic Extraction/PCell library documentation

Calibre/ – directory containing the Calibre verification files

Calibre/XRC/ – directory containing the XRC extraction files

SMIC18MM\_skillUtility – Utility for batch-setting parameter values

## 9. Creation of a Design Project

A unique directory should be created for each circuit design project. The following command can be executed in UNIX:

```
mkdir ~/circuit_design  
cd ~/circuit_design
```

All work by the user should be performed in this circuit design directory. The following file should be copied from the PDK install directory to begin the circuit design process. The following command can be used:

```
cp <pdk_install_directory>/display.drf .
```



Next the user should create a “cds.lib” file. Using any text editor the following entry should be put in the cds.lib file

```
INCLUDE <pdk_install_directory>/cds.lib
```

Where “pdk\_install\_directory” is the path to where the smic0.18um PDK was installed. The following UNIX links are optional but may aid the user in entering certain forms with the Cadence environment. In UNIX the following command can be used:

```
In -s <pdk_install_directory>/models .
```

```
In -s <pdk_install_directory>/stream .
```

Where, again, “pdk\_install\_directory” is the path to where the SMIC PDK was installed.

## 10. Techfile Methodology

The smic 0.18um library techfile will be designed as the master techfile. This techfile will contain all required techfile information. There is an ASCII version of this techfile shipped with the PDK. This ASCII version represents the techfile currently compiled into the smic 0.18um library.

The attach method should be used for any design library that is created. This allows the design database techfile to be kept in sync with the techfile in the process PDK. To create a new library that uses an attached techfile, use the command File->New->Library from either the CIW or library manager and select the “Attach to an existing techfile” option. Select the smic 0.18um library when asked for the name of the Attach to technology library.

## 11. Customizing Layer Display Properties using the display.drf File

The display.drf can be auto loaded at Cadence start-up time or manually loaded during the Cadence session. For the file to be auto loaded, the display.drf file must be located in the



Listed below are the packet, color, lineStyle, and stipplePattern definitions for a metal3 drawing layer. The packet info references predefined color, lineStyle, and stipplePattern definitions. Any of these can be changed to suit an individual user's preferences in the project copy of the display.drf file.

(0 0 0 0 0 0 0 0 0 0 0 0 0 0 0)



## 12. Schematic Design

The user should follow the guidelines listed below while building schematics using Composer: Project libraries should list the primitive PDK library as a reference library in the library properties form.

Users can add instances from the PDK library to designs stored in the project libraries.

When performing hierarchical copy of schematic designs care should be taken to preserve the references to the PDK libraries. These references should not be copied locally to the project directories and the references set to the local copy of PDK cells. This would prevent your designs from inheriting any fixes done to the PDK library from an upgrade.

Users should exercise caution when querying an instance and changing the name of the cell and replacing it with a reference to another cell. While like parameters will inherit values, callbacks are not necessarily executed. This would cause dependent parameters to have incorrect values.

Schematics should be designed with schematic driven layout methodology in mind. Partitioning of schematics, hierarchical design, input and output ports, should be done in a clean and consistent fashion.

Usage of pPar and iPar in a schematic design Contactext is discouraged. While this works fine in schematic design, this could lead to problems while performing schematic driven layout.

## 13. Library Device Setup

### 13.1. MOSFETs

All mosfets in the PDK library are 4 terminal devices, with the body terminal explicitly connected.

Units:

Length and width are in meters, with areas and perimeters in meters squared and meters, respectively. Design variables are allowed for Length and Width entries.

Calculation:

The area and perimeter parameters for the sources and drains are calculated from the width and the number of fingers used. This calculation assumes that the drain will always have the less capacitance (area) when there is an even number of fingers (odd number of diffusion areas). The Width per finger is calculated by dividing the width by the number of fingers. This parameter is for viewing by the designer.

Simulation:

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These mosfets are netlisted as their predefined device names for simulation purposes. SMIC's model definitions are used for these devices.

## 13.2. Resistors

The resistors in the library consist of three types: metal, diffused and insulated. The diffused types include p+, n+, are 2 or 3 terminal resistors with diode back plates. The insulated resistors are those that are isolated from silicon by an insulator (oxide) such as poly resistors. These resistors are 2 or 3 terminal devices. The metal resistors are 2 terminal resistors with the resistive material being the appropriate metal process layer. Serpentine resistor layouts are not allowed.

Units:

The width is specified in meters for schematic simulation. All parameters entered into the resistor form must be integers or floating-point numbers. No design variables are supported due to the calculations that must be performed on the entries.

Calculation:

The length of the resistor segment is calculated from the resistance value, the width, and the numbers of parallel or series segments specified. These calculations are based on the SMIC data provided in the design rule manuals and SPICE models provided by SMIC. See the Foundry Documents in PDK Release Note for document names.

The width and length are snapped to grid, and the resistances are recalculated and updated on the component form based on actual dimensions.

Simulation:

SMIC supplied models are used to model the resistors.

## 13.3. Bipolar Transistors

All BJT's in the PDK library are 3 terminals.

Units:

Only fixed size devices are allowed. A cyclic is used to enter the desired size.

Calculation:

The area is calculated from the emitter size cyclic.

Simulation:

These BJTs are netlisted as their predefined device names for simulation purposes. SMIC's model definitions are used for these devices.



## 13.4. Diodes

All diodes in the PDK library are 2 terminal devices.

Units:

Length and width are in meters. Design variables are not allowed for length and Width entries.

Calculation:

The area is calculated from the width and length entered.

Simulation:

These diodes are netlisted as their predefined device names for simulation purposes. SMIC's model definitions are used for these devices.

## 13.5. Capacitors

Capacitors in SMIC are mimcaps, momcaps and varactors. The mimcaps and momcaps are two terminal metal capacitors and the varactors are two terminal voltage controlled capacitors.

Units:

The length and width of mimcaps are specified in meters for schematic simulation. The finger number and length of momcaps are specified for schematic simulation. The varactors are specified by the number of fingers or groups of unit-varactors. All parameters entered into the capacitor form must be integers or floating-point numbers. No design variables are supported due to the calculations that must be performed on the entries.

Calculation:

The area is calculated from width and length of the mimcaps. And the area is calculated from length and finger number of the momcaps. The appropriate model and number of instances is subsequently netlisted in the simulation environment.

Simulation:

SMIC's model definitions are used for these devices.

## 13.6. Varactors

SMIC provides MOS Varactors and Junction diode Varactors.

All varactors in SMIC PDK are 2 terminal devices.

Units:

The length and width are in meters. Number of fingers must be an integer.

Simulation:

SMIC's model definitions are used for these devices.



## 14. Supported Devices

### 14.1. MOSFETs

| Description                             | Device     | Model Name | Terminal | View                                                                           |
|-----------------------------------------|------------|------------|----------|--------------------------------------------------------------------------------|
| 1.8V nominal VT NMOS transistor         | n18        | n18        | D G S B  | ivpcell,<br>layout,<br>spectre,<br>hspiceD,<br>ams, auLvs,<br>auCdl,<br>symbol |
| 3.3V nominal VT NMOS transistor         | n33        | n33        | D G S B  |                                                                                |
| 1.8V medium VT NMOS transistor          | nmvt18     | nmvt18     | D G S B  |                                                                                |
| 3.3V medium VT NMOS transistor          | nmvt33     | nmvt33     | D G S B  |                                                                                |
| 1.8V native NMOS transistor             | nnt18      | nnt18      | D G S B  |                                                                                |
| 3.3V native NMOS transistor             | nnt33      | nnt33      | D G S B  |                                                                                |
| 1.8V nominal VT PMOS transistor         | p18        | p18        | D G S B  |                                                                                |
| 3.3V nominal VT PMOS transistor         | p33        | p33        | D G S B  |                                                                                |
| 1.8V medium VT PMOS transistor          | pmvt18     | pmvt18     | D G S B  |                                                                                |
| 1.8V Sub-ckt nominal VT NMOS transistor | n18_ckt    | n18_ckt    | D G S B  |                                                                                |
| 3.3V Sub-ckt nominal VT NMOS transistor | n33_ckt    | n33_ckt    | D G S B  |                                                                                |
| 1.8V Sub-ckt medium VT NMOS transistor  | nmvt18_ckt | nmvt18_ckt | D G S B  |                                                                                |
| 3.3V Sub-ckt medium VT NMOS transistor  | nmvt33_ckt | nmvt33_ckt | D G S B  |                                                                                |
| 1.8V Sub-ckt native NMOS transistor     | nnt18_ckt  | nnt18_ckt  | D G S B  |                                                                                |
| 3.3V Sub-ckt native NMOS transistor     | nnt33_ckt  | nnt33_ckt  | D G S B  |                                                                                |
| 1.8V Sub-ckt nominal VT PMOS transistor | p18_ckt    | p18_ckt    | D G S B  |                                                                                |
| 3.3V Sub-ckt nominal VT PMOS transistor | p33_ckt    | p33_ckt    | D G S B  |                                                                                |

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|                                                         |                |                |         |  |
|---------------------------------------------------------|----------------|----------------|---------|--|
| transistor                                              |                |                |         |  |
| 1.8V Sub-ckt medium VT PMOS transistor                  | pmvt18_ckt     | pmvt18_ckt     | D G S B |  |
| 1.8V Sub-ckt DNW nominal VT NMOS transistor             | n18_dnw_ckt    | n18_dnw_ckt    | D G S B |  |
| 4-Terminals 1.8V Sub-ckt DNW nominal VT NMOS transistor | n18_dnw_4t_ckt | n18_dnw_4t_ckt | D G S B |  |
| 3.3V Sub-ckt DNW nominal VT NMOS transistor             | n33_dnw_ckt    | n33_dnw_ckt    | D G S B |  |
| 4-Terminals 3.3V Sub-ckt DNW nominal VT NMOS transistor | n33_dnw_4t_ckt | n33_dnw_4t_ckt | D G S B |  |
| 1.8V DNW Sub-ckt nominal VT PMOS transistor             | p18_dnw_ckt    | p18_dnw_ckt    | D G S B |  |
| 4-Terminals 1.8V Sub-ckt DNW nominal VT PMOS transistor | p18_dnw_4t_ckt | p18_dnw_4t_ckt | D G S B |  |
| 3.3V DNW Sub-ckt nominal VT PMOS transistor             | p33_dnw_ckt    | p33_dnw_ckt    | D G S B |  |
| 4-Terminals Sub-ckt 3.3V DNW nominal VT PMOS transistor | p33_dnw_4t_ckt | p33_dnw_4t_ckt | D G S B |  |
| 3-Terminals 1.8V nominal VT NMOS transistor             | n18_3t         | n18            | D G S   |  |
| 3-Terminals 3.3V nominal VT NMOS transistor             | n33_3t         | n33            | D G S   |  |
| 3-Terminals 1.8V medium VT NMOS transistor              | nmvt18_3t      | nmvt18         | D G S   |  |
| 3-Terminals 3.3V medium VT NMOS transistor              | nmvt33_3t      | nmvt33         | D G S   |  |
| 3-Terminals 1.8V native NMOS transistor                 | nnt18_3t       | nnt18          | D G S   |  |
| 3-Terminals 3.3V native NMOS transistor                 | nnt33_3t       | nnt33          | D G S   |  |
| 3-Terminals 1.8V nominal VT PMOS transistor             | p18_3t         | p18            | D G S   |  |
| 3-Terminals 3.3V nominal VT PMOS transistor             | p33_3t         | p33            | D G S   |  |
| 3-Terminals 1.8V medium VT PMOS transistor              | pmvt18_3t      | pmvt18         | D G S   |  |
| 3-Terminals 1.8V nominal VT NMOS transistor             | n18_ckt_3t     | n18_ckt        | D G S   |  |

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|                                             |               |            |       |  |  |
|---------------------------------------------|---------------|------------|-------|--|--|
| 3-Terminals 3.3V nominal VT NMOS transistor | n33_ckt_3t    | n33_ckt_   | D G S |  |  |
| 3-Terminals 1.8V medium VT NMOS transistor  | nmvt18_ckt_3t | nmvt18_ckt | D G S |  |  |
| 3-Terminals 3.3V medium VT NMOS transistor  | nmvt33_ckt_3t | nmvt33_ckt | D G S |  |  |
| 3-Terminals 1.8V native NMOS transistor     | nnt18_ckt_3t  | nnt18_ckt  | D G S |  |  |
| 3-Terminals 3.3V native NMOS transistor     | nnt33_ckt_3t  | nnt33_ckt  | D G S |  |  |
| 3-Terminals 1.8V nominal VT PMOS transistor | p18_ckt_3t    | p18_ckt    | D G S |  |  |
| 3-Terminals 3.3V nominal VT PMOS transistor | p33_ckt_3t    | p33_ckt    | D G S |  |  |
| 3-Terminals 1.8V medium VT PMOS transistor  | pmvt18_ckt_3t | pmvt18_ckt | D G S |  |  |

## 14.2. Resistors

| Description                                  | Device     | Model Name | Terminal     | View                                                                           |
|----------------------------------------------|------------|------------|--------------|--------------------------------------------------------------------------------|
| N+ diffusion resistor with salicide          | rndif      | rndif      | PLUS MINUS   | ivpcell,<br>layout,<br>spectre,<br>hspiceD,<br>ams, auLvs,<br>auCdl,<br>symbol |
| P+ diffusion resistor with salicide          | rpdif      | rpdif      | PLUS MINUS   |                                                                                |
| N+ diffusion resistor without salicide       | rndifsab   | rndifsab   | PLUS MINUS   |                                                                                |
| P+ diffusion resistor without salicide       | rpdifsab   | rpdifsab   | PLUS MINUS   |                                                                                |
| N+ poly resistor with salicide               | rnpo       | rnpo       | PLUS MINUS   |                                                                                |
| 3-terminal N+ poly resistor with salicide    | rnpo_3t    | rnpo_3t    | PLUS MINUS B |                                                                                |
| P+ poly resistor with salicide               | rppo       | rppo       | PLUS MINUS   |                                                                                |
| 3-terminal P+ poly resistor with salicide    | rppo_3t    | rppo_3t    | PLUS MINUS B |                                                                                |
| N+ poly resistor without salicide            | rnposab    | rnposab    | PLUS MINUS   |                                                                                |
| 3-terminal N+ poly resistor without salicide | rnposab_3t | rnposab_3t | PLUS MINUS B |                                                                                |
| P+ poly resistor without salicide            | rpposab    | rpposab    | PLUS MINUS   |                                                                                |
| 3-terminal P+ poly resistor without salicide | rpposab_3t | rpposab_3t | PLUS MINUS B |                                                                                |
| 1K High resistance poly resistor             | rhypo      | rhypo      | PLUS MINUS   |                                                                                |
| 1K 3-terminal High resistance poly           | rhypo_3t   | rhypo_3t   | PLUS MINUS B |                                                                                |

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|                                                        |                |                |              |
|--------------------------------------------------------|----------------|----------------|--------------|
| resistor                                               |                |                |              |
| Nwell resistor under AA                                | rnwaa          | rnwaa          | PLUS MINUS   |
| Nwell resistor under STI                               | rnwsti         | rnwsti         | PLUS MINUS   |
| Metal resistors                                        | rmx            | rmx            | PLUS MINUS   |
| Thick Top metal resistor(22K)                          | rtm2           | rtm2           | PLUS MINUS   |
| Thick Top metal resistor(34K)                          | rmtt1          | rmtt1          | PLUS MINUS   |
| Thick Top metal resistor(40K)                          | rmtt2          | rmtt2          | PLUS MINUS   |
| N+ diffusion resistor with salicide subckt model       | rndif_ckt      | rndif_ckt      | PLUS MINUS   |
| P+ diffusion resistor with salicide subckt model       | rpdif_ckt      | rpdif_ckt      | PLUS MINUS   |
| N+ diffusion resistor without salicide subckt model    | rndifsab_ckt   | rndifsab_ckt   | PLUS MINUS   |
| P+ diffusion resistor without salicide subckt model    | rpdiffsab_ckt  | rpdiffsab_ckt  | PLUS MINUS   |
| N+ poly resistor with salicide subckt model            | rnpo_ckt       | rnpo_ckt       | PLUS MINUS   |
| 3-terminal N+ poly resistor with salicide subckt model | rnpo_3t_ckt    | rnpo_3t_ckt    | PLUS MINUS B |
| P+ poly resistor with salicide subckt model            | rppo_ckt       | rppo_ckt       | PLUS MINUS   |
| 3-terminal P+ poly resistor with salicide subckt model | rppo_3t_ckt    | rppo_3t_ckt    | PLUS MINUS B |
| N+ poly resistor without salicide subckt model         | rnposab_ckt    | rnposab_ckt    | PLUS MINUS   |
| 3-terminal N+ poly resistor without salicide subckt    | rnposab_3t_ckt | rnposab_3t_ckt | PLUS MINUS B |
| P+ poly resistor without salicide subckt model         | rpposab_ckt    | rpposab_ckt    | PLUS MINUS   |
| 3-terminal P+ poly resistor without salicide subckt    | rpposab_3t_ckt | rpposab_3t_ckt | PLUS MINUS B |
| High resistance poly resistor subckt model             | rhypo_ckt      | rhypo_ckt      | PLUS MINUS   |
| 3-terminal High resistance poly resistor subckt model  | rhypo_3t_ckt   | rhypo_3t_ckt   | PLUS MINUS B |
| Nwell resistor under AA subckt model                   | rnwaa_ckt      | rnwaa_ckt      | PLUS MINUS   |
| Nwell resistor under STI subckt model                  | rnwsti_ckt     | rnwsti_ckt     | PLUS MINUS   |

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### 14.3. Bipolar Transistors

| Description                 | Device        | Model Name        | Terminal | View                                                                  |
|-----------------------------|---------------|-------------------|----------|-----------------------------------------------------------------------|
| 3.3V PNP bipolar transistor | pnp33         | pnp33a4           | E B C    | ivpcell, layout,<br>spectre, hspiceD,<br>ams, auLvs, auCdl,<br>symbol |
|                             |               | pnp33a25          | E B C    |                                                                       |
|                             |               | pnp33a100         | E B C    |                                                                       |
| 1.8V PNP bipolar transistor | pnp18         | pnp18a4           | E B C    |                                                                       |
|                             |               | pnp18a25          | E B C    |                                                                       |
|                             |               | pnp18a100         | E B C    |                                                                       |
| 3.3V NPN bipolar transistor | nnp33         | nnp33a4           | E B C    |                                                                       |
|                             |               | nnp33a25          | E B C    |                                                                       |
|                             |               | nnp33a100         | E B C    |                                                                       |
| 1.8V NPN bipolar transistor | nnp18         | nnp18a4           | E B C    |                                                                       |
|                             |               | nnp18a25          | E B C    |                                                                       |
|                             |               | nnp18a100         | E B C    |                                                                       |
| 3.3V PNP bipolar transistor | pnp33_mis_ckt | pnp33a4_mis_ckt   | E B C    |                                                                       |
|                             |               | pnp33a25_mis_ckt  | E B C    |                                                                       |
|                             |               | pnp33a100_mis_ckt | E B C    |                                                                       |
| 1.8V PNP bipolar transistor | pnp18_mis_ckt | pnp18a4_mis_ckt   | E B C    |                                                                       |
|                             |               | pnp18a25_mis_ckt  | E B C    |                                                                       |
|                             |               | pnp18a100_mis_ckt | E B C    |                                                                       |
| 3.3V NPN bipolar transistor | nnp33_mis_ckt | nnp33a4_mis_ckt   | E B C    |                                                                       |
|                             |               | nnp33a25_mis_ckt  | E B C    |                                                                       |
|                             |               | nnp33a100_mis_ckt | E B C    |                                                                       |
| 1.8V NPN bipolar transistor | nnp18_mis_ckt | nnp18a4_mis_ckt   | E B C    |                                                                       |

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|  |  |                   |       |  |
|--|--|-------------------|-------|--|
|  |  | npn18a25_mis_ckt  | E B C |  |
|  |  | npn18a100_mis_ckt | E B C |  |

## 14.4. Diodes

| Description                          | Device    | Model Name | Terminal   | ivpcell, layout, spectre,<br>hspiceD, ams, auLvs,<br>auCdl, symbol |
|--------------------------------------|-----------|------------|------------|--------------------------------------------------------------------|
| 1.8V N+/Pwell diode                  | ndio18    | ndio18     | PLUS MINUS |                                                                    |
| 1.8V P+/Nwell diode                  | pdio18    | pdio18     | PLUS MINUS |                                                                    |
| 3.3V N+/Pwell diode                  | ndio33    | ndio33     | PLUS MINUS |                                                                    |
| 3.3V P+/Nwell diode                  | pdio33    | pdio33     | PLUS MINUS |                                                                    |
| Nwell/P substrate diode              | nwdio     | nwdio      | PLUS MINUS |                                                                    |
| 1.8V native N+/Pwell diode           | nndio18   | nndio18    | PLUS MINUS |                                                                    |
| 3.3V native N+/Pwell diode           | nndio33   | nndio33    | PLUS MINUS |                                                                    |
| Buried Pwell/ Deep Nwell diode       | diobpw    | diobpw     | PLUS MINUS |                                                                    |
| 1.8V MVT N+ to P-Well junction diode | ndiomvt18 | ndiomvt18  | PLUS MINUS |                                                                    |
| 1.8V MVT P+ to N-Well junction diode | pdiomvt18 | pdiomvt18  | PLUS MINUS |                                                                    |
| 3.3V MVT N+ to P-Well junction diode | ndiomvt33 | ndiomvt33  | PLUS MINUS |                                                                    |

## 14.5. Capacitors

| Description                                                                    | Device   | Model Name | Terminal   | View                                                          |
|--------------------------------------------------------------------------------|----------|------------|------------|---------------------------------------------------------------|
| Topmetal-MiM capacitor, with $C_{spec}=1\text{ fF}/\mu\text{m}^2$              | mim      | mim        | PLUS MINUS | ivpcell,<br>layout,<br>spectre,<br>hspiceD,<br>ams,<br>auLvs, |
| Topmetal-MiM capacitor, with $C_{spec}=1\text{ fF}/\mu\text{m}^2$ subckt model | mim_ckt  | mim_ckt    | PLUS MINUS |                                                               |
| Topmetal-MiM capacitor, with $C_{spec}=2\text{ fF}/\mu\text{m}^2$              | mim2     | mim2       | PLUS MINUS |                                                               |
| Topmetal-MiM capacitor, with $C_{spec}=2\text{ fF}/\mu\text{m}^2$              | mim2_ckt | mim2_ckt   | PLUS MINUS |                                                               |

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|              |  |  |  |                  |
|--------------|--|--|--|------------------|
| subckt model |  |  |  | auCdl,<br>symbol |
|--------------|--|--|--|------------------|

## 14.6. Varactors

| Description                        | Device     | Model Name | Terminal   | View                                                               |
|------------------------------------|------------|------------|------------|--------------------------------------------------------------------|
| 1.8V N+ poly/Nwell MOS<br>Varactor | pvar18_ckt | pvar18_ckt | PLUS MINUS | ivpcell, layout, spectre,<br>hspiceD, ams, auLvs, auCdl,<br>symbol |
| 3.3V N+ poly/Nwell MOS<br>Varactor | pvar33_ckt | pvar33_ckt | PLUS MINUS |                                                                    |

## 15. CDF parameters

### 15.1. MOSFETs

Multiplier – number of Parallel MOS devices

Model Name – spectre model name (non-editable)

Length (M) – gate length in meters

Total Width (M) – gate width in meters (sum of all fingers)

Finger Width – width of each gate finger/stripe

Fingers – number of poly gate fingers/stripes used in layout

Threshold – finger width at which to apply device folding of the layout

Apply Threshold –button to apply threshold or not

Gate Connection –allow shorting of multi-finger devices and addition of contact heads to gate end.

CT enc by S/D AA (horizontal) – enclosure of S/D AA to its contact along channel length direction. This value will impact the LOD parameter (SA/SB).

Multi CTs enc by S/D AA (vertical) – enclosure of S/D AA to its contact along channel width direction. When contact number =1, the value will be a fixed number calculated by channel width and contact size. When contact numbers >=2, user can input value at design requirement, a SKILL callback will ensure the input value to be in an allowable range. This parameter may impact contact numbers and contact space, but will not impact channel width.

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S/D contact to gate spacing – distance between S/D contact and channel gate. This value will impact the LOD parameter (SA/SB).

Switch S/D – source is defined as left-most diffusion region and alternating regions to the right. Pins are not automatically permuted and can be switched using this parameter.

Bodytie Type – None, Detached, or Integrated (butting source).

- For Detached, user may select Left, Right, Top, and/or Bottom to specify the located of bodyties. Selection of all four creates a guardring.
- For Detached, the user may specify Tap Extension(in microns) which sets the distance from the bodytie to the device. Maximum distance is 100 microns.
- For Integrated, the user may select Left or Right for a device with an odd number of fingers(1,3,5,). The user may select Left and Right for an even fingered device. This option does not apply to native and medium Vth devices.

Drain diffusion area, Source diffusion area, Drain diffusion periphery, Source diffusion periphery – several simulation parameters, which are calculated and netlisted in accordance with SMIC models.

Source/Drain diffusion res squares – Approximately calculated the number of resistor squares in source area and drain area, this is for pre-simulation. LVS/RCX will extract the more precise number from real layout for post-simulation.

Edit LOD Parameter – the value can be “false” or “true”. It is an editable switch of “Left Dist. OD & poly (SA)” / “Right Dist. OD & poly (SB)” / “Gate Spacing (SD)”

- For “false”, the “Left Dist. OD & poly (SA)” / “Right Dist. OD & poly (SB)” / “Gate Spacing (SD)” will be calculated by “CT enc by S/D AA(horizontal)” and “S/D contact to gate spacing” automatically.
- For “true”, the “Left Dist. OD & poly (SA)” / “Right Dist. OD & poly (SB)” / “Gate Spacing (SD)” can be input by the user.

Left Dist. OD & poly (SA) / Right Dist. OD & poly (SB) – parameters for LOD effect simulation, indicates the distance between STI and channel gate. The value can be a calculated number or an input number which controlled by the switch of “Edit LOD Parameter”.

Gate Spacing (SD) – Parameter for LOD effect simulation, indicates distance between channel gates when finger numbers  $\geq 2$ . When finger number =1, the value will be 0, when finger numbers  $\geq 2$ , the value can be calculated number by parameter “S/D CT to gate spacing” or input by user.

SCA/SCB/SCC – WPE parameter for simulation, which are reserved parameters for future model, and are set to 0 in current version PDK.

Mismatch model flag – 2 options are provided: “1” turns on mismatch simulation, “0” turns it off. Default is set to “1”



**Add Instance**

Hide Cancel Defaults Help

Library: smic18nmrf Browse

Cell: n18

View: symbol

Names:

Array Rows: 1 Columns: 1

Rotate Sideways Upside Down

Model Name: n18

Multiplier: 1

Length: 180n M

Total Width: 220n M

Finger Width: 220n M

Fingers: 1

Threshold: 220n M

Apply Threshold: ☐

Gate Connection: ☒ None ☐ Top ☐ Bottom

Gate Direction: ☒ Mid ☐ Left ☐ Right

S/D AA vertical enc to multi: 100n M

CT enc by horizontal S/D AA: 100n M

S/D CT to gate spacing: 200n M

Switch S/D: ☐

Bodytie Type: None

Edit Area & Perim: ☐

Drain diffusion area: 198.4f M

Source diffusion area: 198.4f M

Figure15-1 MOSFET ADD INSTANCE FORM



## 15.2. Resistors

Model name (model) – Spectre model name (non-editable)

Multiplier (m) – number of parallel resistor devices.

Segments (segments) – Number of resistor segments

Segment Connection (connection) – Connection between segments (Parallel, Series)

Calculated Parameter (calculatedParam) – Parameter to be calculated (Resistance, Length)

Resistance (r) – Entry or calculated value for resistance.

Segment Width (segW) – Entry value for segment width in meters.

Segment Length (segL) – Entry or calculated value for segment Length in meters, not including interface resistance.

Effective Width (effW) – Calculated value based on number of segments, segment width and segment connection (non-editable)

Effective Length (effL) – Calculated value based on number of segments, segment width, and segment connection (non-editable)

Subckt resistor calculated Width (cktW) – Calculated value based on segment width.

Subckt resistor calculated Length (cktL) – Calculated value for sub-circuit resistor length (total resistance including the interface resistance).

Left Dummy (leftDummy) – Toggle for placement of dummy resistor segment on the left side

Right Dummy (rightDummy) – Toggle for placement of dummy resistor segment on the right side

Contact Rows (cntRows) – Entry value for number of contact rows in each segment head

Contact Columns (cntCols) – Calculated value for number of contact columns in each segment head (non-editable)

3Terminal Tap Flag – Allow user to select well taps for the 3rd terminal connection of poly resistor. Options are “Nwell”, “Pwell” and “None”. Default value is “None”. Only exists in 3-T Poly Resistors

Mismatch model flag – 2 options are provided: “1” turns on mismatch simulation, “0” turns it off. Default is set to “1”

Flag\_CC - 2 options are provided: “1” turns on mismatch simulation, “0” turns it off. Default

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is set to "1"

**Add Instance**

Hide Cancel Defaults Help

Library: smic18nmrf Browse

Cell: rposab\_3t\_ckt

View: symbol

Names:

Array Rows: 1 Columns: 1

Rotate Sideways Upside Down

Model name: rposab\_3t\_ckt

Multiplier: 1

Segments: 1

Segment Connection: Series

Calculated Parameter: Resistance

Resistance: 7.65337K Ohms

Segment Width: 500n M

Segment Length: 10u M

Effective Width: 500n M

Effective Length: 10u M

Subckt resistor calculated W: 500n M

Subckt resistor calculated L: 10u M

Left Dummy: ☐

Right Dummy: ☐

Contact Rows: 1

Contact Columns: 1

3Terminals Tap Flag: None

Figure15-2 RESISTOR ADD INSTANCE FORM



## 15.3. Bipolar Transistors

Model name – Spectre model name (non-editable)

Emitter Size – cyclic of the “width x length” of the emitter (microns).

Area – calculated emitter area based on emitter width & length in meters squared.

Multiplier – Number of Parallel Bipolar devices

Mismatch model flag – 2 options are provided: “1” turns on mismatch simulation, “0” turns it off. Default is set to “1”

Figure15-3

## 15.4. Diodes

Model name – Spectre model name (non-editable)

Device Area – Calculated junction area in meters squared (non-editable)

Width (M) – Entry value for diode width in meters.

Length (M) – Entry value for diode length in meters.



Multiplier – Number of Parallel Diode devices

Periphery of junction – Calculated junction periphery in meters (non-editable).

Figure15-4 DIODE ADD INSTANCE FORM

## 15.5. Capacitors

Model name – Spectre model name (non-editable)

Multiplier – Number of Parallel capacitor devices.

Calculate – Parameter to be calculated (Capacitance, Length, Width, width & length).

Total Capacitance – Calculated capacitance based on width, length and multiplier.

Capacitance – Calculated capacitance based on width and length in Farad.

Width (M) – Entry value for capacitor width in meters.

Length (M) – Entry value for capacitor length in meters.

Mismatch model flag – 2 options are provided: “1” turns on mismatch simulation, “0” turns it off. Default is set to “1”



Figure15-5 CAPACITOR ADD INSTANCE FORM

## 15.6. Varactors

Model name – Spectre model name (non-editable)

Multiplier – Number of Parallel varactor devices.

Fingers – Number of poly gate fingers or AA stripes used in layout.

Width (M) – Varactor width in meters.

Length (M) – Varactor length in meters.

Voltage difference – voltage input by user for capacitance calculation.

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Capacitance – Calculated capacitance based on w/l and voltage and multiplier under different Voltage.

Mismatch model flag – 2 options are provided: “1” turns on mismatch simulation, “0” turns it off. Default is set to “1”

Figure15-6 VARACTOR ADD INSTANCE FORM

## 16. Component Label Defaults

### 16.1. MOSFETs

- component parameters: model, l, w, m

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- operating point: ids, vgs, vds, vth, vdsat
- model: vto, kp, gamma
- instance name prefix: NM, PM

## 16.2. Resistors

- component parameters: r, w, l, segments, connection Type
- operating point: v, i, res
- model:
- instance name prefix: R, X

## 16.3. Bipolar Transistors

- component parameters: model, area, m
- operating point: betadc, ic, Vce
- model: bf, is, va
- instance name prefix: Q

## 16.4. Diodes

- component parameters: model, area, m
- operating point: id, vd, reg
- model: is, rs, n
- instance name prefix: D

## 16.5. Capacitors

- component parameters: c, w, l, m
- operating point: cap
- model:
- instance name prefix: C



## 16.6. Varactors

- component parameters: w, l, nf, m
- operating point:
- model:
- instance name prefix: C

## 17. Simulation Models

The following simulators are supported in this PDK:

Spectre  
hspiceD  
AMS

The following model library setup for Spectre is done automatically when the user first opens the SMIC PDK library or uses a component from that library. The user may disable this feature by editing the libInit.il file found in the SMIC PDK library.

```
<pdtk_install_directory>/models/spectre/xxxx_spe.lib    tt
<pdtk_install_directory>/models/spectre/xxxx_spe.lib    res_tt
<pdtk_install_directory>/models/hspice/xxxx.lib          TT
<pdtk_install_directory>/models/hspice/xxxx.lib          RES_TT
```

Where <pdtk\_install\_directory> is the path where the SMIC PDK library is installed, and the second entry is the simulation corner that the user wants to simulate.

The user should follow the instructions provided with the device models from SMIC to ensure the proper selection of sections for simulation.

## 18. Add Wire Utility

Certain devices in the library have bulk pins (i.e. n12, p33, etc.). A utility has been added to the PDK to automatically wire these bulk terminals to user specified signals (i.e. gnd!, vdd!, etc.). This will help reduce schematic clutter while maintaining required circuit hookup. The following picture shows the result of executing the smic13mm\_addWire routine.

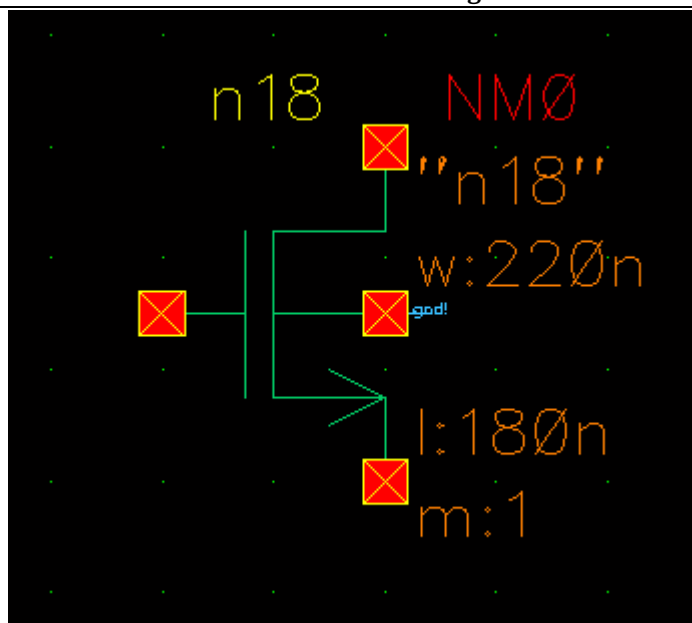


Figure18-1 Schematic

There are no arguments to run the program, the user must type `smic18mm_addWire()` in the CIW. If you don't have a schematic cellview open it will give you an error message.

If the user has instances selected when they run `addWire`, it will prompt them for the label name for those specific instances and will only add the wires to those instances. It does nothing to the schematic hierarchy.

If no instance is selected then the program will wire up all instances that have their bulk nodes unconnected. The user gets prompted for the label name for each type of instance, and they also have the option of running it down through the hierarchy or just at current cellview.

And, finally, if the user does not want to use `gnd!` or `vdd!` as the label name, there is an entry box for the user to type in an alternative net name. If another wire name issued, it will be added to the cyclic list of label name choices - but only for that DfII session. Once you exit the dfill session, the cyclic is reset to `gnd!` and `vdd!`.

## 19. Utility

### 19.1. Update CDFs Utility

When changes are made to a device inside the PDK, these changes often affect circuit design which has already been created. For example, a sheet resistance `Rsh` value may change on a resistor which affects the length of the device passed to the simulator and used to generate the layout.



These parameters are not automatically updated in each of the designer's circuit libraries. A function has been written and included as a part of the PDK to complete the update to an existing library such that all modifications made to a PDK since a previous release are reflected in each of the circuit designs inside a library.

The procedure name as typed in the CIW is PasFlexUpdateLib. Executing the command in CIW brings up a GUI for updating design libraries. After specifying a design library, the procedure traverses the design library hierarchy and updates all the appropriate CDF information.

Please note, however, that possible LVS violations may arise as a result of running this routine depending upon what changes have been made to the PDK. For example, a sheet resistance Rsh change as specified earlier could cause a resistor to shrink in length in the schematic thus causing a mismatch as far as LVS is concerned. Please be sure that you verify again each design in simulation, DRC, and LVS to insure that no unintended modifications have been overlooked.

## 19.2. Batch setting CDF values Utility

A utility is provided to set CDF values for a certain schematic design in a batch mode. Currently only mis-match flag parameter is supported (mismod / mismod\_res / mismod\_res\_rf / mismod\_mim / mismod\_mim\_rf), and only works on 'schematic' view.

Execute the utility by loading the skill file under  
"<pdk\_library>/SMIC18MMRF\_skillUtility/batchSetCDF.ile"

## 20. Techfile Layers

Techfile layers defined in the PDK (techfile.tf) are done in compliance with technology file supplied by SMIC techfile group. The technology file compiled with this PDK contains all defined layer/levels as defined in the SMIC techfile group supplied techfile. Layer names and visibility /selectability may be altered from this technology file, if necessary.

## 21. Virtuoso XL

The standard Cadence Virtuoso XL design flow is implemented. This includes basic connectivity of connection layers, wells, and substrate, and symbolic contacts. The M factor is used for device instance multiplier - there is no conflict with the parameter used in cell

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operation. Names are displayed on the layout views to aid in schematic-layout instance correlation. Auto-abutment of MOSFET devices is supported. Pin permuting of MOSFET and resistor devices is also supported. The pcell layouts are compiled into the PDK.

The users should follow the guidelines listed below for layout design:

- The VirtuosoXL tool requires a separate license for operation.
- Users obtain maximum leverage from the PDK by doing schematic driven layout in the Virtuoso XL environment. This flow will produce a correct design layout. The Virtuoso Custom Router (IC Craftsman) can be used to finish the interconnection in the layout.
- The VCR rules file for the target process is provided with the PDK (icc.rules).
- Abutment is currently supported only for MOS transistors. Note, abutment will work only on schematic driven layouts.
- Schematic Driven Layout is recommended over Netlist Driven Layout.

NOTE: Skill pcell source code is not included in the PDK kit.

## 21.1. Symbolic Contacts

|        |                                   |
|--------|-----------------------------------|
| M1_AA  | Metal-1 to Active Area Contact    |
| M1_GT  | Metal-1 to Poly Contact           |
| M1_NW  | Metal-1 to Nwell Contact          |
| M1_SN  | Metal-1 to N+ Active Area Contact |
| M1_SP  | Metal-1 to P+ Active Area Contact |
| M1_SUB | Metal-1 to P+ Substrate Contact   |
| M2_M1  | Metal-2 to Metal-1 Via            |
| M3_M2  | Metal-3 to Metal-2 Via            |
| M4_M3  | Metal-4 to Metal-3 Via            |
| M5_M4  | Metal-5 to Metal-4 Via            |
| M6_M5  | Metal-6 to Metal-5 Via            |
| M7_M6  | Metal-7 to Metal-6 Via            |
| M8_M7  | Top-Metal to Metal-7 Via          |

CREATE CONTACT FORM

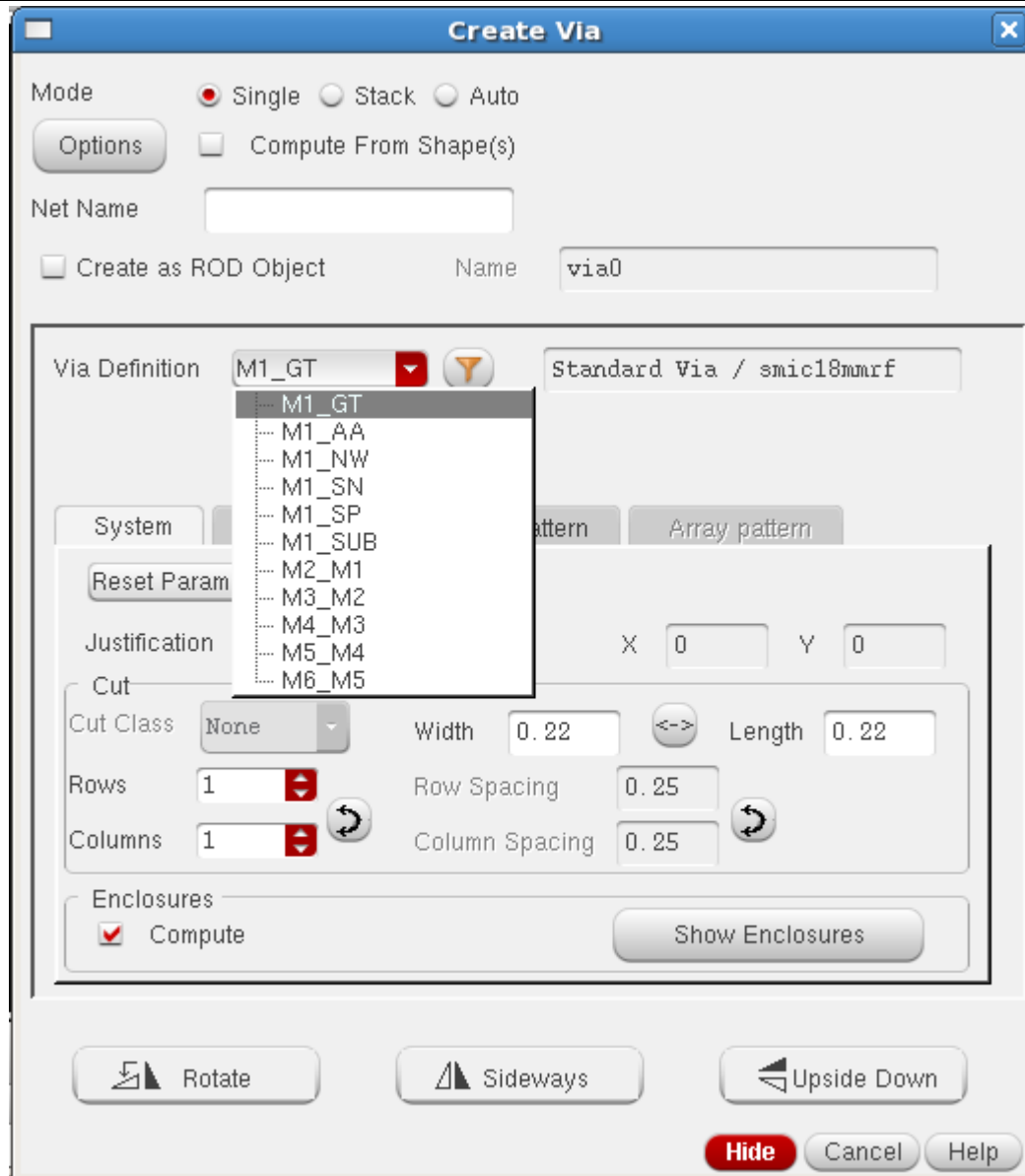


Figure21-1 CREATE CONTACT FORM

## 22. Known Problems & Solutions

- 1) There may be M1 area DRC violation in base cell layout like mosfets ...etc for S/D connection or pin connection; this error would disappear when designer connects this metal out in real design.
- 2) There may be M1/Mn/SN... etc. minimum area DRC violations for symbolic. These errors would disappear in circuit level design.
- 3) LVS may not pass for DNW MOS if the CDF "mDnwGR" is enable. To avoid it, Please

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designer should draw NWell ring manually if insists to enable the CDF "rmDnwGR".

4) There may be CT.2 DRC violation when certain size mos abutting. User can adjust the Gate connection direction to avoid violations.

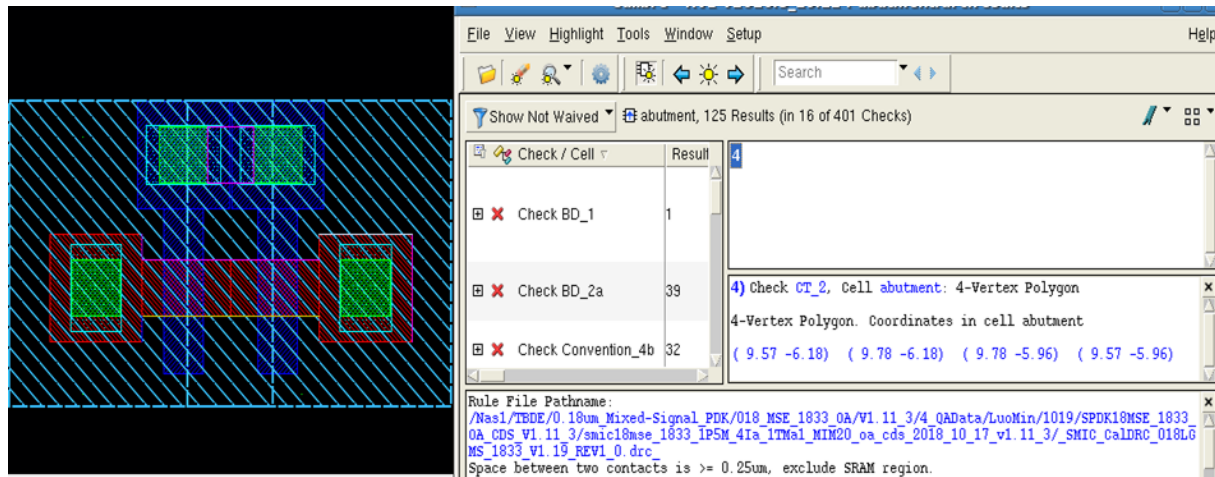


Figure22-1

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